

Lab 4 ticket

Predict a System Through Time — Friday, October 23

Name: _____ Language (circle): Octave / Python

New syntax (fill as you go)

I needed to...	Command / pattern (your language)
Enter A and x_0	
One step $x \leftarrow Ax$	
Loop many steps	
Plot component vs time	

Quiz 7 today is Week 8 systems / RREF (**not** this lab). Week 13 returns to today's mystery.

E1. $A = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}$, $x_0 = (1000, 0)$. Hand x_1, x_2 ; check in code.

Expected: $x_1 = (900, 100)$; $x_2 = (830, 170)$. **Yours:** $x_1 =$ _____ $x_2 =$ _____

E2. Same A , x_0 : loop 20 steps. Record x_5 and x_{20} (ok).

Expected: $x_5 \approx (722.7, 277.3)$; $x_{20} \approx (666.9, 333.1)$. **Yours:** _____

E3. Plot both components vs $t = 0..20$. Pattern name: _____

Expected: stabilization.

E4. What-if: $A_{\text{new}} = \begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix}$, same x_0 . Hand x_1 ; note $t = 20$ first component vs E2.

Expected: $x_1 = (700, 300)$; ~ 400 at $t = 20$ (lower than E2). **Yours:** $x_1 =$ _____ $t = 20$: _____

One-sentence compare: _____

E5. Mystery: $B = \text{diag}(1.2, 0.5)$. Simulate from $(1, 0)$ and from $(0, 1)$.

From $(1, 0)$: _____ From $(0, 1)$: _____

My Week-13 question: _____

Stuck on (optional): E _____

Instructor initials: _____